Full-Dimensional Landmark Solve

Landmark-Driven Depth Control for Sketch-Based Inflation — math notes, compared with the original Monster Mash inflation.
Code: `src/reconstruction.cpp` (Monster Mash), `src/landmarksolver.cpp` (ours).

Notation

Symbol	Meaning
$V,\ F$	mesh vertices and triangles; x, y come from the sketch
L	cotangent Laplacian of the *flat* mesh ($z = 0$); negative semi-definite (libigl convention)
M	diagonal Voronoi mass matrix; M_{ii} is the area around vertex i
∂	outline (boundary) vertices
z or u	depth *before* the square root (the "membrane" field)
h	final height after the square root
p	pressure (right-hand side of the Poisson equation)

1. Original Monster Mash inflation

Step 1 — Flat operators

$$L = \text{cotmatrix}(V_{\text{flat}}, F), \qquad M = \text{massmatrix}(V_{\text{flat}}, F)$$

Step 2 — One constant pressure per part

$$p_v = c_i \ \text{ for every vertex } v \text{ in part } i, \qquad c_i = 2 \ (\text{default}), \qquad c_{\text{back}} = -\,c_{\text{front}}$$

Step 3 — Poisson solve with the outline clamped

$$M^{-1} L\, z = p \ \text{ on interior vertices}, \qquad z = 0 \ \text{ on } \partial$$

This is the discrete form of $\Delta z = p$. It is solved once and is linear in p.

Step 4 — Square-root profile

$$h = \text{sgn}(z)\,\sqrt{|z|}$$

Afterwards the parts are merged and every coordinate is scaled by the subsampling factor $s = 2$. For a disk of radius R with uniform c, $|z| = c\,(R^2 - r^2)/4$ and therefore $h = \frac{\sqrt{c}}{2}\sqrt{R^2 - r^2}$: an exact circular profile.

> Monster Mash has a few degrees of freedom — one pressure value per part — and no depth control inside a part.

2. Full-dimensional landmark solve

Step 0 — Input

The rest mesh produced by Monster Mash, V_0 with heights h_0, and the landmarks.

Step 1 — Same flat operators

L and M are built on V_0 flattened to $z = 0$, exactly as in Monster Mash.

Step 2 — Undo the square root

$$u_0 = \operatorname{sgn}(h_0)\, h_0^2 \qquad \big(= s^2\, z_{\mathrm{MM}}\big)$$

The default pressure is implicit: $p_0 = M^{-1} L\, u_0 \approx s^2 c_i$ inside each part. The constant s^2 is irrelevant because everything is linear.

Step 3 — Unknown: a correction to the default

$$u = u_0 + \Delta u, \qquad \Delta u = 0 \ \text{ on } \partial \quad (\text{outline stays clamped})$$

Because $L\, u = M\, p$ is linear, the corresponding pressure change is

$$\Delta p = M^{-1} L\, \Delta u \qquad (\text{one value per vertex: the full representation}).$$

Step 4 — Energy: smallest pressure change

$$E(\Delta u) \;=\; \tfrac{1}{2}\, \Delta p^{\top} M\, \Delta p \;=\; \tfrac{1}{2}\, \Delta u^{\top} \underbrace{L^{\top} M^{-1} L}_{Q}\, \Delta u$$

This discretizes $\frac{1}{2} \int (\Delta\, \Delta u)^2\, dA$. Away from landmarks the minimizer satisfies $\Delta^2 \Delta u = 0$: the correction is **biharmonic** (as in Zhang et al. 2001 and Botsch & Kobbelt 2004, discretized as in Jacobson et al. 2010).

Alternative (smoothest pressure change, triharmonic): $Q_B = L^{\top} M^{-1} (-L)\, M^{-1} L$ — smoother but less local.

Step 5 — Landmark constraints

Landmark k lies in triangle f_k with barycentric weights β_k; row b_k of the sparse matrix B holds those three weights. The default membrane value at the landmark is

$$\bar{u}_k = b_k^{\top} u_0.$$

A height factor f_k means $h \mapsto f_k\, h$. Since $u \propto h^2$, the target and the residual are

$$t_k = f_k^2\, \bar{u}_k, \qquad r_k = t_k - b_k^{\top} u_0 = \big(f_k^2 - 1\big)\, \bar{u}_k.$$

Stacking all landmarks gives $B\,\Delta u = r$. An anchor is the same constraint with its factor frozen; $f_k = 1$ gives $r_k = 0$ exactly.

Step 6a — Hard constraints ($w = \infty$)

$$\min_{\Delta u} \ \tfrac{1}{2}\,\Delta u^\top Q\,\Delta u \quad \text{s.t.} \quad B\,\Delta u = r, \quad \Delta u_\partial = 0$$

With interior unknowns I and Lagrange multipliers λ, this is the KKT system

$$\begin{bmatrix} Q_{II} & B_I^\top \\ B_I & 0 \end{bmatrix} \begin{bmatrix} \Delta u_I \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}.$$

Step 6b — Soft constraints (weight w)

$$\min_{\Delta u} \ \tfrac{1}{2}\,\Delta u^\top Q\,\Delta u + \tfrac{w}{2}\,\left\| B\,\Delta u - r \right\|^2 \quad \Longrightarrow \quad \left(Q + w\,B^\top B\right)_{II} \Delta u_I = w\,B_I^\top\, r$$

As $w \to \infty$ this becomes the hard case; as $w \to 0$, $\Delta u \to 0$ and the default Monster Mash shape is recovered.

Step 7 — Back to heights

$$u = u_0 + \Delta u, \qquad h = \operatorname{sgn}(u)\,\sqrt{|u|}, \qquad x,\ y \text{ unchanged}$$

Step 8 — Why dragging is live

The system matrix depends only on *where* the landmarks are (and on w); the right-hand side depends only on the factors f_k. Adding or removing a landmark, or changing w, costs one factorization (LU for the KKT system, Cholesky for the soft case); each drag step is a single back-substitution (about 1 ms on 5k vertices). Equivalently (representer theorem), the hard solution blends one response per landmark:

$$\Delta u = \sum_k \lambda_k\, g_k, \qquad g_k = Q^{-1} b_k, \qquad \lambda = \left(B\,Q^{-1}B^\top\right)^{-1} r,$$

where each g_k is a biharmonic Green's function centred at landmark k, with zero depth change on the outline.

3. Comparison

	Monster Mash	**Full landmark solve**
Pressure	constant per part, c_i	$p_0 + \Delta p$, one value per vertex
How it is chosen	fixed default (2)	smallest Δp meeting the landmark targets
Equation	Poisson: $M^{-1}L\,z = p$	Poisson for u_0 + biharmonic correction Δu
Unknown	z	Δu on top of Monster Mash's result
Boundary	$z = 0$ on the outline	$\Delta u = 0$ on the outline
User input	none (regions, layering)	factors f_k, anchors, weight w
Linear solves	one per reconstruction	one factorization per landmark change, one back-substitution per drag
Square root	$h = \operatorname{sgn}(z)\sqrt{\lvert z\rvert}$	same, after undoing it: $u_0 = \operatorname{sgn}(h_0)\,h_0^2$
No landmarks	—	$\Delta u = 0$: exactly Monster Mash

Global-constant baseline. Restricting $\Delta p = (\alpha - 1)\,p_0$ (one degree of freedom) gives $u = \alpha\,u_0$, so every height scales by $\sqrt{\alpha}$. This is the one-dimensional representation the full solve is compared against in RQ1.

4. Caveats of the current formulation

1. **Interpolation mismatch.** Constraints act on the interpolated u, while the displayed height inside a triangle is the interpolated $\sqrt{\cdot}$; $\sqrt{b_k^\top u} \neq b_k^\top \sqrt{u}$ in general. They agree exactly at vertices.
2. **No positivity.** Biharmonic solutions have no maximum principle and can overshoot. Conflicting nearby landmarks can push u through zero, after which the square root flips part of the sheet (the likely cause of the observed fold). A fix is an inequality constraint such as $\operatorname{sgn}(u_0)\,u \geq 0$ or $p \geq 0$, which turns the solve into a QP.
3. **Outline rows couple front and back.** $Q = L^\top M^{-1} L$ also penalizes Δp at outline vertices, whose Laplacian rows sum neighbours on both sheets, so a front edit can slightly affect the back near the silhouette. Dropping or treating those rows separately is the boundary-condition choice discussed by Stein et al. 2018.

References

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Reference venues were listed from memory; verify before citing.